

LAMBERTS BAY NORTIER, South Africa

WMO: 686130

Lat: 32.035S

Lon: 18.332E

Elev: 94

StdP: 100.21

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.7	6.6	-3.1	3.0	20.3	-1.2	3.5	19.8	8.9	17.7	7.7	17.0	2.7	100	0.419

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.6	32.5	18.2	29.9	17.4	27.8	16.8	20.5	26.8	19.7	25.1	19.0	23.9	5.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.7	21.7	17.9	13.0	21.0	17.3	12.5	20.6	59.2	26.9	56.5	25.3	54.4	24.0	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.7	7.5	6.6	DB	4.0	37.4	1.0	1.7	3.3	38.6	2.7	39.6	2.2	40.6	1.4	41.8
			WB	3.3	22.0	1.0	0.6	2.6	22.4	2.0	22.8	1.5	23.1	0.8	23.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	19.6	19.9	19.0	18.0	15.7	14.0	13.5	13.1	14.7	16.3	17.4	18.6
(7)		DBStd	3.78	2.01	2.49	2.66	4.04	3.54	3.08	3.68	2.81	3.14	3.11	2.36	2.07
(8)		HDD10.0	6	0	0	0	0	0	1	4	1	0	0	0	0
(9)		HDD18.3	913	7	6	18	53	103	139	159	166	119	80	44	19
(10)		CDD10.0	2428	298	278	279	242	176	120	113	98	142	196	221	265
(11)		CDD18.3	293	46	51	39	44	21	8	10	5	11	17	16	26
(12)		CDH23.3	2016	216	315	276	367	148	42	49	36	108	173	134	153
(13)	CDH26.7	682	60	117	94	138	46	8	6	7	44	67	43	52	
(14)	Wind	WSAvg	3.2	3.3	3.1	3.1	3.0	2.9	3.2	3.2	3.1	3.2	3.4	3.5	3.3
(15)	Precipitation	PrecAvg	219	5	3	6	15	29	35	41	34	19	9	10	7
(16)		PrecMax	346	22	15	19	36	90	72	133	71	54	40	47	39
(17)		PrecMin	118	0	0	0	2	0	0	7	10	1	0	0	0
(18)		PrecStd	61	7	4	5	10	23	18	32	19	13	9	11	9
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.6	35.2	33.9	34.5	31.3	28.0	27.2	27.4	31.5	33.5	30.7	32.8
(20)			MCWB	19.0	19.4	18.7	18.6	16.3	14.9	14.7	14.1	15.8	17.5	16.7	18.6
(21)		2%	DB	28.1	30.7	29.3	30.4	27.4	24.1	24.5	23.2	26.0	28.2	27.2	27.1
(22)			MCWB	18.3	18.9	18.3	16.8	14.9	13.5	13.3	13.2	13.7	16.0	16.3	17.4
(23)		5%	DB	25.3	27.3	26.4	27.5	24.3	21.3	22.0	20.3	22.2	24.6	23.9	24.3
(24)			MCWB	18.0	18.2	17.5	16.3	14.0	12.6	12.4	11.8	13.2	14.9	15.7	17.5
(25)		10%	DB	23.7	24.6	23.9	24.7	21.2	18.9	19.4	17.4	19.3	21.6	21.8	22.8
(26)			MCWB	18.0	18.0	17.6	15.4	13.8	12.0	11.7	12.0	13.1	14.5	15.9	17.3
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.1	21.7	21.3	20.0	17.4	16.1	15.7	15.8	16.6	19.1	19.5	20.3
(28)			MADB	27.6	29.5	28.0	31.8	26.1	22.3	23.0	22.0	25.5	29.1	24.5	26.4
(29)		2%	WB	20.0	20.5	20.0	18.3	16.2	14.8	14.4	14.5	15.3	17.1	18.2	19.2
(30)			MADB	24.4	27.0	25.5	27.3	22.5	19.3	20.0	19.5	21.2	23.3	22.4	24.2
(31)		5%	WB	19.2	19.5	19.1	17.2	15.5	14.1	13.6	13.6	14.6	16.2	17.3	18.5
(32)			MADB	23.4	24.8	23.8	24.0	20.5	17.6	18.2	17.5	19.3	21.1	21.4	22.7
(33)		10%	WB	18.6	18.7	18.3	16.4	14.8	13.5	12.8	12.9	14.0	15.5	16.6	17.8
(34)			MADB	22.5	23.1	22.4	22.0	18.7	16.7	16.9	15.9	17.7	19.8	20.6	21.5
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.3	8.6	8.4	8.9	8.1	7.9	8.7	8.4	8.8	9.0	8.4	8.4
(36)			MADB	11.0	12.7	12.3	13.3	13.9	12.2	12.6	13.5	13.6	13.7	12.5	10.8
(37)			MCWBR	4.5	4.7	4.6	4.9	5.2	5.1	5.4	5.9	5.5	5.8	5.2	4.7
(38)		5% WB	MADB	8.3	9.1	8.8	10.3	9.9	8.4	9.6	9.3	9.9	9.6	8.6	8.7
(39)			MCWBR	4.3	4.8	4.7	4.7	5.1	5.1	5.5	5.6	5.4	5.2	4.8	4.6
(40)	Clear-Sky Solar Irradiance	taub		0.334	0.336	0.331	0.316	0.308	0.300	0.304	0.323	0.333	0.342	0.327	0.332
(41)		taud		2.544	2.546	2.556	2.596	2.599	2.587	2.567	2.511	2.453	2.448	2.537	2.539
(42)		Ebn at Noon		1005	983	955	918	875	857	869	894	936	965	1005	1010
(43)		Edh at Noon		109	106	98	84	75	71	76	90	107	115	110	111
(44)	All-Sky Solar Radiation	RadAvg		8.32	7.53	6.19	4.62	3.33	2.85	3.17	3.93	5.28	6.77	7.93	8.47
(45)		RadStd		0.22	0.20	0.18	0.23	0.21	0.15	0.16	0.18	0.23	0.28	0.25	0.33

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.04	N/A	N/A	N/A	N/A

Nomenclature: See separate page